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NEET · CHEMISTRY · CHAPTER-WISE PYQS

Some Basic Concepts of Organic Chemistry

NEET Chemistry · Previous Year Questions with Answer Key

127	49	1988 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	126 questions
MCQ (More than One Correct Answer)	1 question
Answer key	at the end

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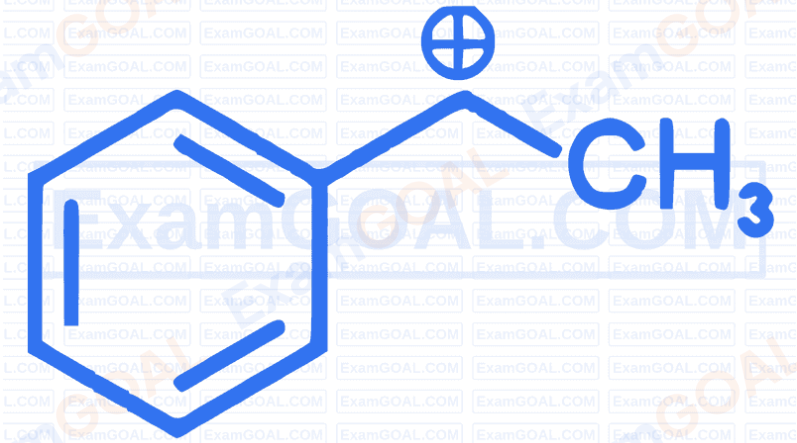
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MCQ (Single Correct Answer) 126 questions

Q1. RE-NEET 2026

The following carbocation is stabilized by the interaction of the empty p orbital with

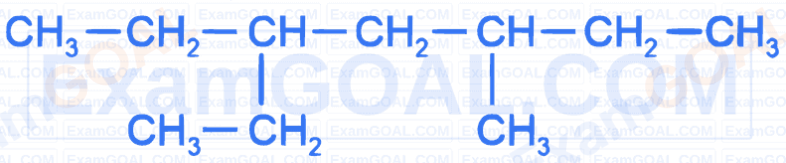


Open solution

(A) empty σ^* and empty π^* orbitals
(B) filled σ and filled π orbitals
(C) empty σ and empty π^* orbitals
(D) empty σ^* and filled π orbitals

Q2. NEET 2026

The correct IUPAC name of the following compound is :



Open solution

(A) 3-ethyl-5-methylheptane
(B) 2,4-diethylhexane
(C) 3-methyl-5-ethylheptane
(D) 3,5-diethylhexane

Q3. NEET 2026

During Lassaigne's test, the elements present in an organic compound are converted from :

Open solution

(A) Ionic form to ionic form
(B) Covalent form to ionic form
(C) Covalent form to covalent form
(D) Ionic form to covalent form

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Q4. NEET 2025

How many products (including stereoisomers) are expected from monochlorination of the following compound?



Open solution

(A) 5 (B) 6 (C) 2 (D) 3

Q5. NEET 2025

Total number of possible isomers (both structural as well as stereoisomers) of cyclic ethers of molecular formula C_4H_8O is :

Open solution

(A) 10 (B) 11 (C) 6 (D) 8

Q6. NEET 2025

Among the given compounds I-III, the correct order of bond dissociation energy of C-H bond marked with * is :



Open solution

(A) III > II > I (B) II > III > I (C) II > I > III (D) I > II > III

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Q7. NEET 2025

Open solution

Match List-I with List-II

	List - I (Mixture)		List - II (Method of separation)
A	$CHCl_3 + C_6H_5NH_2$	I	Distillation under reduced pressure
B	Crude oil in petroleum industry	II	Steam distillation
C	Glycerol from spent-lye	III	Fractional distillation
D	Aniline - water	IV	Simple distillation

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II
(B) A-III, B-IV, C-II, D-I
(C) A-IV, B-III, C-I, D-II
(D) A-IV, B-III, C-II, D-I

Q8. NEET 2025

Open solution

Which one of the following compounds can exist as cis-trans isomers?

(A) 1, 1-Dimethylcyclopropane
(B) 1, 2-Dimethylcyclohexane
(C) Pent-1-ene
(D) 2-Methylhex-2-ene

Q9. NEET 2025

Open solution

Match List-I with List-II

	List - I (Ion)		List - II (Group Number in Cation Analysis)
A	Co^{2+}	I	Group-I
B	Mg^{2+}	II	Group-III
C	Pb^{2+}	III	Group-IV
D	Al^{3+}	IV	Group-VI

Choose the correct answer from the options given below :

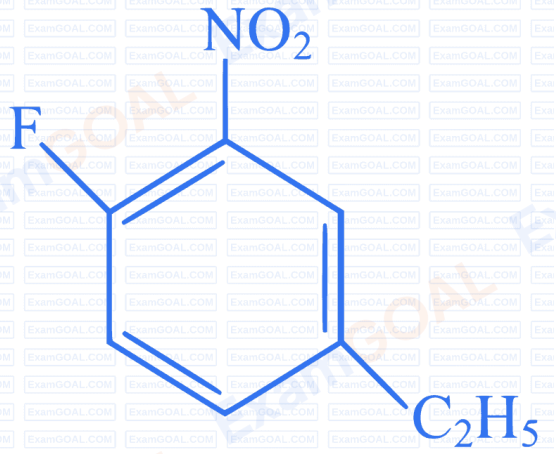
(A) A-III, B-II, C-IV, D-I
(B) A-III, B-II, C-I, D-IV
(C) A-III, B-IV, C-II, D-I
(D) A-III, B-IV, C-I, D-II

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Q10. NEET 2024 (Re-Examination)

Open solution

The correct IUPAC name of the compound



is :

(A) 4-ethyl-1-fluoro-2-nitrobenzene
(B) 4-ethyl-1-fluoro-6-nitrobenzene
(C) 3-ethyl-6-fluoro-1-nitrobenzene
(D) 1-ethyl-4-fluoro-3-nitrobenzene

Q11. NEET 2024 (Re-Examination)

Open solution

A steam volatile organic compound which is immiscible with water has a boiling point of $250^{\circ}C$. During steam distillation, a mixture of this organic compound and water will boil :

(A) above $100^{\circ}C$ but below $250^{\circ}C$
(B) above $250^{\circ}C$
(C) at $250^{\circ}C$
(D) close to but below $100^{\circ}C$

Q12. NEET 2024 (Re-Examination)

Open solution

Match List-I with List-II :

	List - I (Test/reagent)		List - II (Radical identified)
(A)	Lake Test	(I)	NO_3^-
(B)	Nessler's Reagent	(II)	Fe^{3+}
(C)	Potassium sulphocyanide	(III)	Al^{3+}
(D)	Brown Ring Test	(IV)	NH_4^+

Choose the correct answer from the options given below :

(A) A-IV, B-II, C-III, D-I
(B) A-II, B-IV, C-III, D-I
(C) A-II, B-III, C-IV, D-I
(D) A-III, B-IV, C-II, D-I

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Q13. NEET 2024 (Re-Examination)

Open solution

Methyl group attached to a positively charged carbon atom stabilizes the carbocation due to

(A) –I inductive effect
(B) electromeric effect
(C) hyperconjugation
(D) mesomeric effect

Q14. NEET 2024

Open solution

Given below are two statements:

Statement I : The boiling point of three isomeric pentanes follows the order

n-pentane > isopentane > neopentane

Statement II : When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct
(B) Both Statement I and Statement II are incorrect
(C) Statement I is correct but Statement II is incorrect
(D) Statement I is incorrect but Statement II is correct

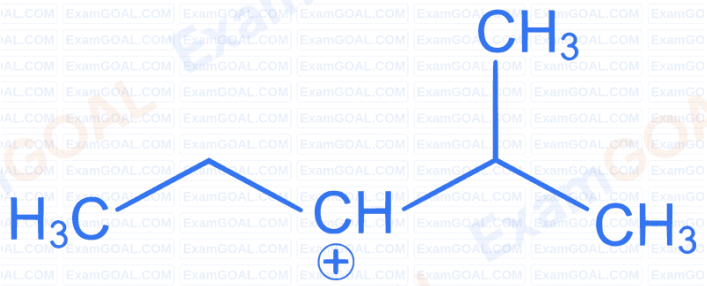
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Q15. NEET 2024

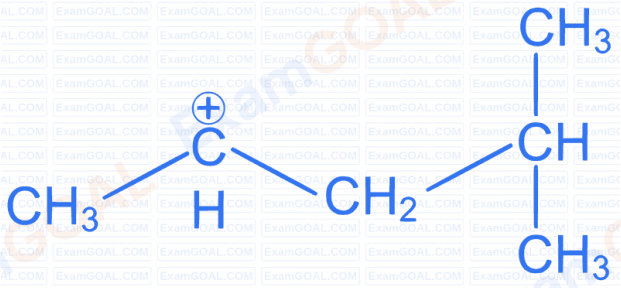
The most stable carbocation among the following is :

Open solution

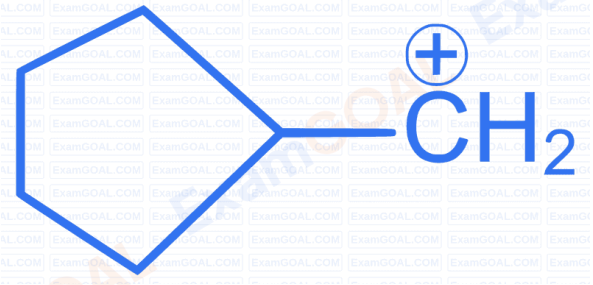
(A)



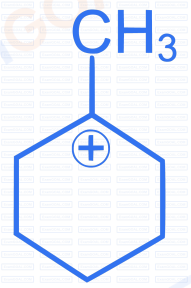
(B)



(C)



(D)



Q16. NEET 2024

A compound with a molecular formula of C_6H_{14} has two tertiary carbons. Its IUPAC name is :

Open solution

(A) n-hexane

(B) 2-methylpentane

(C) 2,3-dimethylbutane

(D) 2,2-dimethylbutane

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Q17. NEET 2024

On heating, some solid substances change from solid to vapour state without passing through liquid state. The technique used for the purification of such solid substances based on the above principle is known as

Open solution

(A) Crystallization (B) Sublimation (C) Distillation (D) Chromatography

Q18. NEET 2024

During the preparation of Mohr's salt solution (Ferrous ammonium sulphate), which of the following acid is added to prevent hydrolysis of Fe^{2+} ion?

Open solution

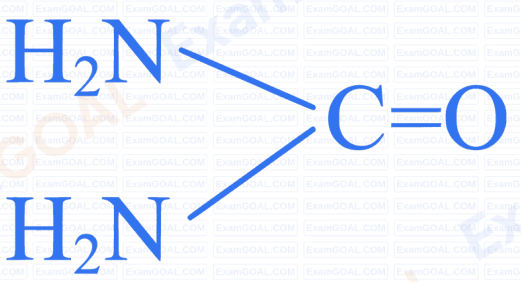
(A) dilute hydrochloric acid
(B) concentrated sulphuric acid
(C) dilute nitric acid
(D) dilute sulphuric acid

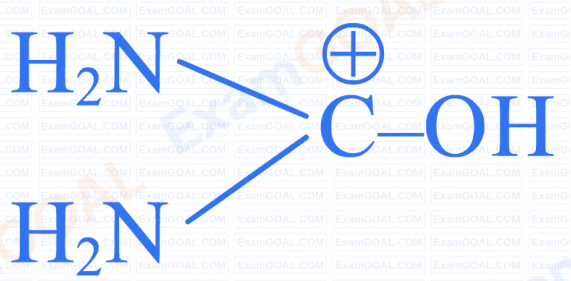
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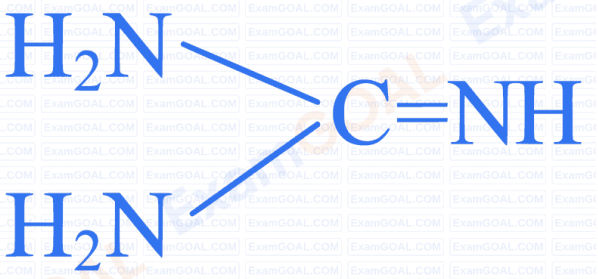
Q19. NEET 2023 Manipur

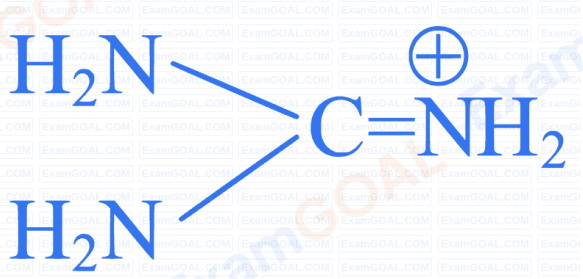
Which amongst the following compounds/species is least basic?

Open solution

(A) 

(B) 

(C) 

(D) 

Q20. NEET 2023 Manipur

Which amongst the following compounds will show geometrical isomerism?

Open solution

(A) Pent-1-ene

(B) 2, 3-Dimethylbut-2-ene

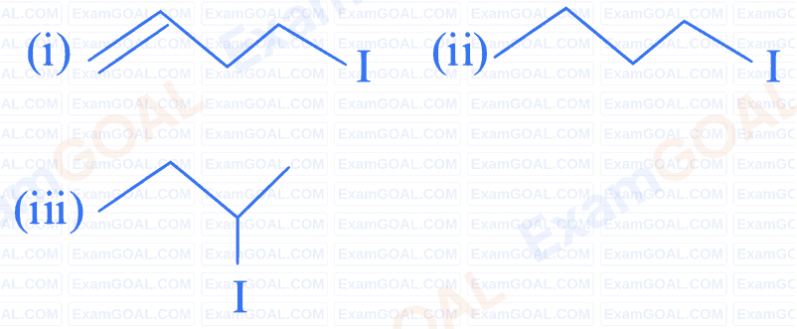
(C) 2-Methylprop-1-ene

(D) 3, 4-Dimethylhex-3-ene

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Q21. NEET 2023 Manipur

The correct order for the rate of α, β-dehydrohalogenation for the following compounds is _____.



Open solution

(A) (i) < (ii) < (iii)

(B) (ii) < (i) < (iii)

(C) (iii) < (ii) < (i)

(D) (ii) < (iii) < (i)

Q22. NEET 2023 Manipur

Given below are two statements:

Statement I : In an organic compound, when inductive and electromeric effects operate in opposite directions, the inductive effect predominates.

Statement II : Hyperconjugation is observed in o-xylene.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement-I is true but Statement-II is false.

(B) Statement-I is false but Statement-II is true.

(C) Both Statement-I and Statement-II are true.

(D) Both Statement-I and Statement-II are false.

Open solution

Q23. NEET 2023

The number of σ bonds, π bonds and lone pair of electrons in pyridine, respectively are:

(A) 12, 3, 0

(B) 11, 3, 1

(C) 12, 2, 1

(D) 11, 2, 0

Open solution

Q24. NEET 2023

In Lassaigne's extract of an organic compound, both nitrogen and sulphur are present, which gives blood red colour with Fe^{3+} due to the formation of-

(A) NaSCN

(B) $[Fe(CN)_5NOS]^{4-}$

(C) $[Fe(SCN)]^{2+}$

(D) $Fe_4[Fe(CN_6)]_3 \cdot xH_2O$

Open solution

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Q25. NEET 2023

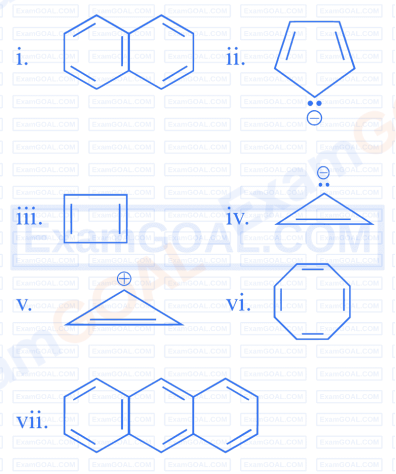
Amongst the given options which of the following molecules/ion acts as a Lewis acid?

(A) H_2O (B) BF_3 (C) OH^- (D) NH_3

Open solution

Q26. NEET 2023

Consider the following compounds/species:



The number of compounds/species which obey Huckel's rule is _____.

(A) 6 (B) 2 (C) 5 (D) 4

Open solution

Q27. NEET 2022 Phase 2

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Chlorine is an electron withdrawing group but it is ortho, para directing in electrophilic aromatic substitution.

Reason (R) : Inductive effect of chlorine destabilises the intermediate carbocation formed during the electrophilic substitution, however due to the more pronounced resonance effect, the halogen stabilises the carbocation at ortho and para positions.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) (A) is not correct but (R) is correct

(B) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(C) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(D) (A) is correct but (R) is not correct

Open solution

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Q28. NEET 2022 Phase 2

Open solution

Predict the order of reactivity of the following four isomers towards S_N2 reaction.

(I) $CH_3CH_2CH_2CH_2Cl$

(II) $CH_3CH_2CH(Cl)CH_3$

(III) $(CH_3)_2CHCH_2Cl$

(IV) $(CH_3)_3CCl$

(A) (IV) > (II) > (III) > (I)

(B) (IV) > (III) > (II) > (I)

(C) (I) > (II) > (III) > (IV)

(D) (I) > (III) > (II) > (IV)

Q29. NEET 2022 Phase 2

Open solution

What is the hybridization shown by C_1 and C_2 carbons, respectively in the given compound?

$OHC - CH = CH - CH_2COOCH_3$

(A) sp^3 and sp^3

(B) sp^2 and sp^3

(C) sp^2 and sp^2

(D) sp^3 and sp^2

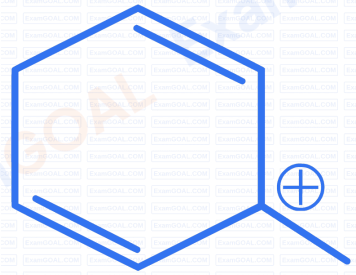
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Q30. NEET 2022 Phase 2

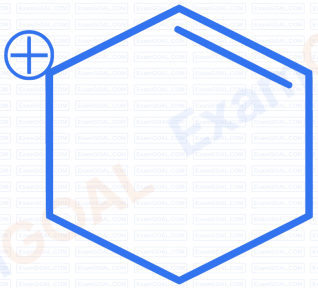
Open solution

Which of the following is the most stable carbocation?

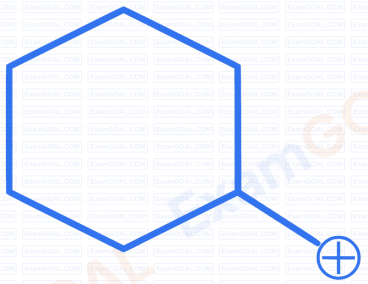
(A)



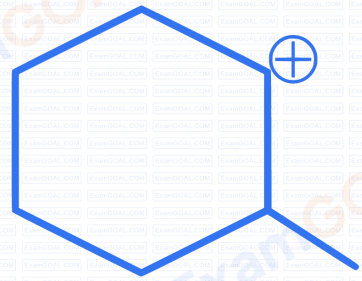
(B)



(C)



(D)



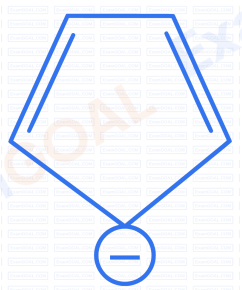
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Q31. NEET 2022 Phase 1

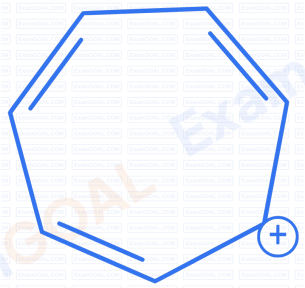
Which compound amongst the following is not an aromatic compound?

Open solution

(A)



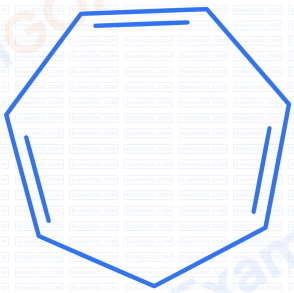
(B)



(C)



(D)



Q32. NEET 2022 Phase 1

The incorrect statement regarding chirality is

Open solution

(A) S_N1 reaction yields 1 : 1 mixture of both enantiomers

(B) The product obtained by S_N2 reaction of haloalkane having chirality at the reactive site shows inversion of configuration

(C) Enantiomers are superimposable mirror images on each other

(D) A racemic mixture shows zero optical rotation

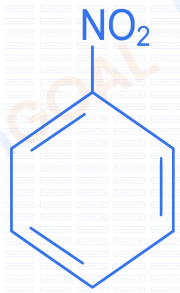
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Q33. NEET 2022 Phase 1

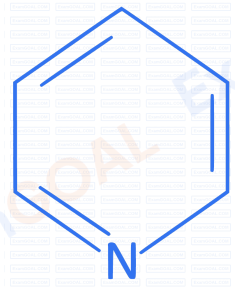
The Kjeldahl's method for the estimation of nitrogen can be used to estimate the amount of nitrogen in which one of the following compounds?

Open solution

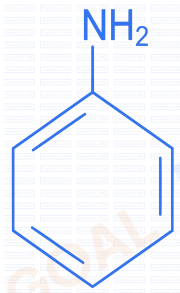
(A)



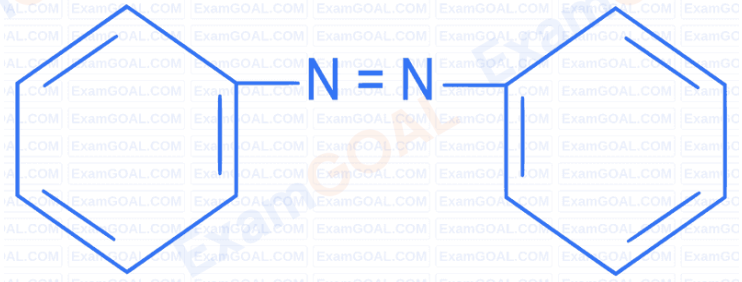
(B)



(C)



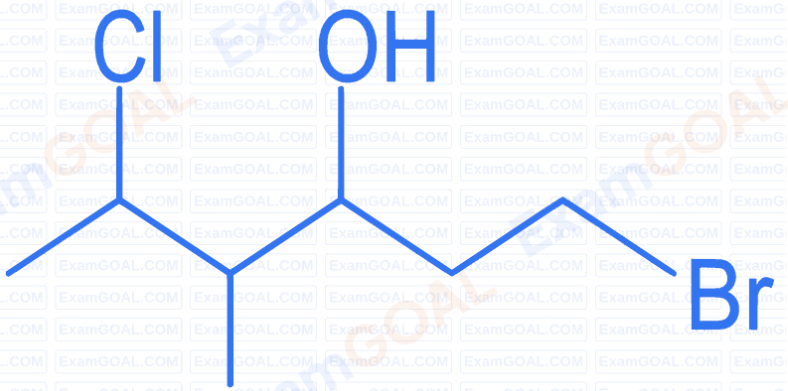
(D)



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Q34. NEET 2022 Phase 1

The correct IUPAC name of the following compound is



Open solution

(A) 1-bromo-5-chloro-4-methylhexan-3-ol
(B) 6-bromo-2-chloro-4-methylhexan-4-ol
(C) 1-bromo-4-methyl-5-chlorohexan-3-ol
(D) 6-bromo-4-methyl-2-chlorohexan-4-ol

Q35. NEET 2021

The compound which shows metamerism is :

(A) $C_4H_{10}O$
(B) C_5H_{12}
(C) C_3H_8O
(D) C_3H_6O

Open solution

Q36. NEET 2021

Dihedral angle of least stable conformer of ethane is :

(A) 0°
(B) 120°
(C) 180°
(D) 60°

Open solution

Q37. NEET 2020 Phase 2

A liquid compound (x) can be purified by steam distillation only if it is

(A) Not steam volatile, immiscible with water
(B) Steam volatile, immiscible with water
(C) Not steam volatile, miscible with water
(D) Steam volatile, miscible with water

Open solution

Q38. NEET 2020 Phase 1

Paper chromatography is an example of :

(A) Partition chromatography
(B) Thin layer chromatography
(C) Column chromatography
(D) Adsorption chromatography

Open solution

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Q39. NEET 2020 Phase 1

A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of which of the following?

Open solution

(A) +R effect of $-CH_3$ groups

(B) $-$ R effect of $-CH_3$ groups

(C) Hyperconjugation

(D) $-$ I effect of $-CH_3$ groups

Q40. NEET 2019 (Odisha)

The most stable carbocation, among the following, is

Open solution

(A) $CH_3-CH_2-\overset{\oplus}{C}H_2$

(B) $(CH_3)_3C-\overset{\oplus}{C}H-CH_3$

(C) $CH_3-CH_2-\overset{\oplus}{C}H-CH_2-CH_3$

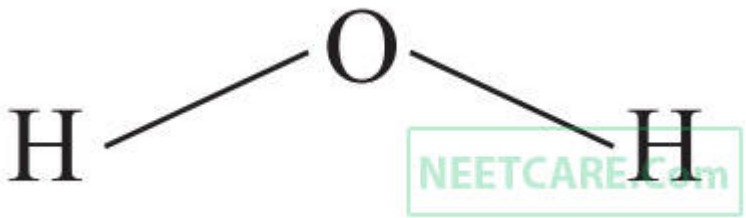
(D) $CH_3-\overset{\oplus}{C}H-CH_2-CH_2-CH_3$

Q41. NEET 2019

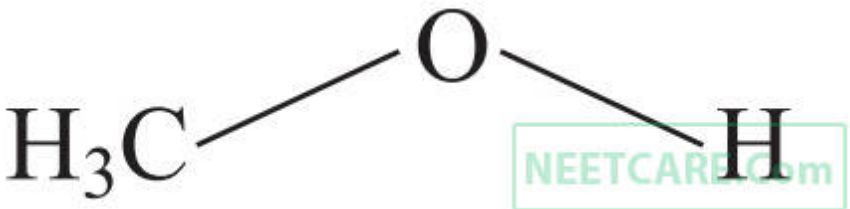
The compound that is most difficult to protonate is :

Open solution

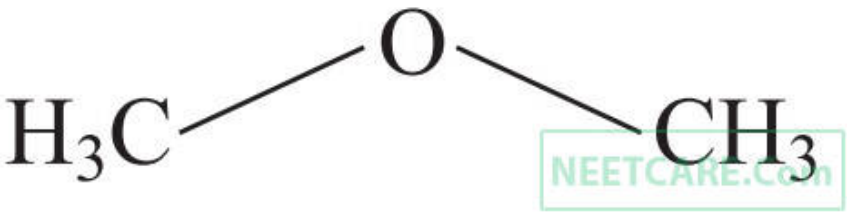
(A)



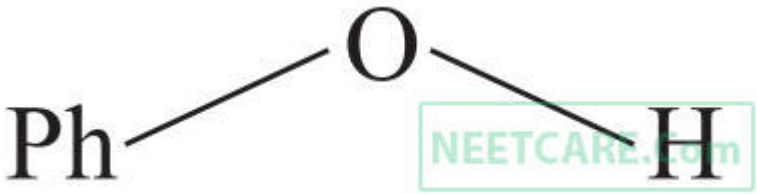
(B)



(C)



(D)



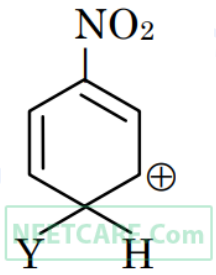
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Q42. NEET 2018

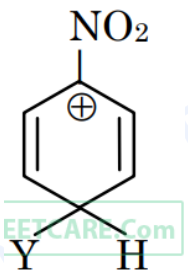
Which of the following carbocations is expected to be most stable?

Open solution

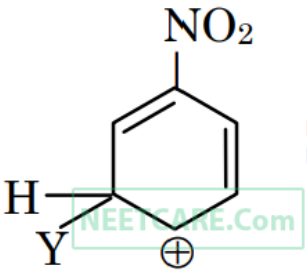
(A)



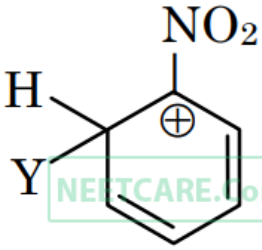
(B)



(C)



(D)



Q43. NEET 2018

Which of the following molecules represents the order of hybridisation sp^2 , sp^2 , sp, sp from left to right atoms?

Open solution

(A) $HC \equiv C - C \equiv CH$

(B) $CH_2 = CH - C \equiv CH$

(C) $CH_2 = CH - CH = CH_2$

(D) $CH_3 - CH = CH - CH_3$

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Q44. NEET 2017

The most suitable method of separation of 1 : 1 mixture of ortho and para-nitrophenols is

(A) chromatography
(B) crystallisation
(C) steam distillation
(D) sublimation.

Open solution

Q45. NEET 2017

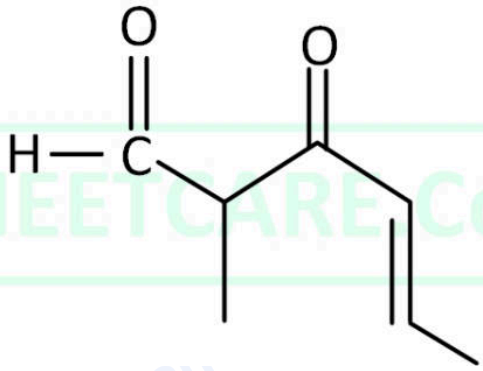
The correct statement regarding electrophile is

(A) electrophile is a negatively charged species and form a bond by accepting a pair of electrons from another electrophile
(B) electrophiles are generally neutral species and can form a bond by accepting a pair of electrons from a nucleophile
(C) electrophile can be either neutral or positively charged species and can form a bond by accepting a pair of electrons from a nucleophile
(D) electrophile is a negatively charged species and can form a bond by accepting a pair of electrons from a nucleophile.

Open solution

Q46. NEET 2017

The IUPAC name of the compound



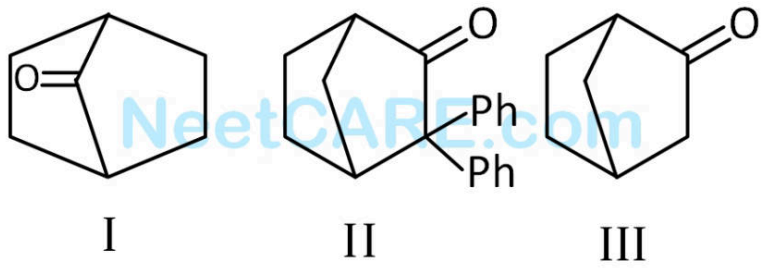
(A) 5-formylhex-2-en-3-one
(B) 5-methyl-4-oxohex-2-en-5-al
(C) 3-keto-2-methylhex-5-enal
(D) 3-keto-2-methylhex-4-enal

Open solution

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Q47. NEET 2016 Phase 2

Which among the given molecules can exhibit tautomerism?



I II III

(A) III only (B) Both I and III (C) Both I and II (D) Both II and III

Open solution

Q48. NEET 2016 Phase 1

The correct statement regarding the comparison of staggered and eclipsed conformations of ethane, is

(A) the eclipsed conformation of ethane is more stable than staggered conformation even though the eclipsed conformation has torsional strain

(B) the staggered conformation of ethane is more stable than eclipsed conformation, because staggered conformation has no torsional strain

(C) the staggered conformation of ethane is less stable than eclipsed conformation, because staggered conformation has torsional strain

(D) the eclipsed conformation of ethane is more stable than staggered conformation, because eclipsed conformation has no torsional strain.

Open solution

Q49. AIPMT 2015

Which of the following statements is not correct for a nucleophile?

(A) Ammonia is a nucleophile.

(B) Nucleophiles attack low e^- density sites.

(C) Nucleophiles are not electron seeking.

(D) Nucleophile is a Lewis acid.

Open solution

Q50. AIPMT 2015

The possible stereo-structures of $CH_3CHOHCOOH$, which are optically active, are called

(A) atropisomers (B) enantiomers (C) mesomers (D) diastereomers

Open solution

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Q51. AIPMT 2015 Cancelled Paper

The total number of π-bond electrons in the following structure is

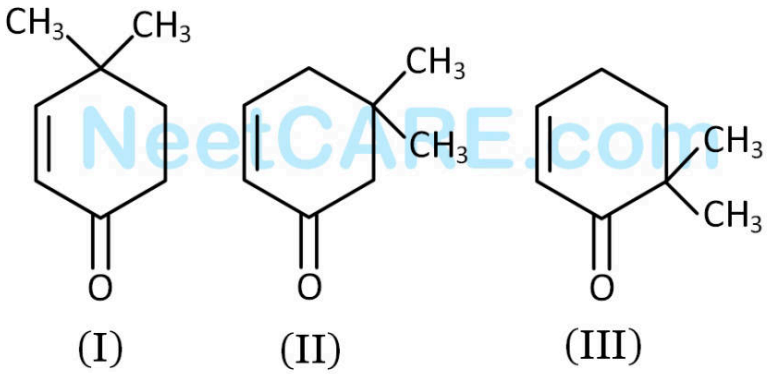


Open solution

(A) 12 (B) 16 (C) 4 (D) 8

Q52. AIPMT 2015 Cancelled Paper

Given :



(I) (II) (III)

Open solution

Which of the given compounds can exhibit tautomerism?

(A) II and III (B) I, II and III (C) I and II (D) I and III

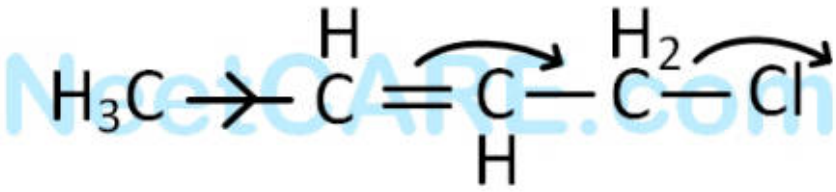
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Q53. AIPMT 2015 Cancelled Paper

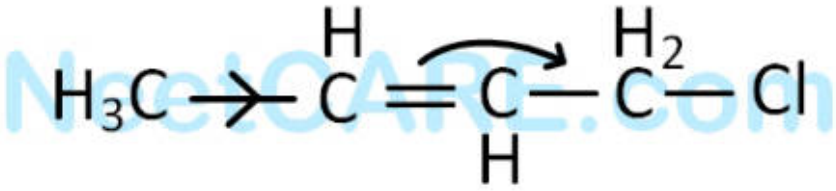
Which of the following is the most correct electron displacement for a nucleophilic reaction to take place?

Open solution

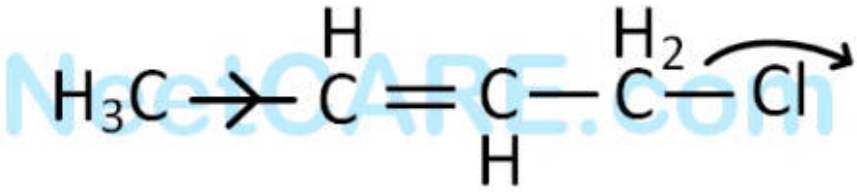
(A)



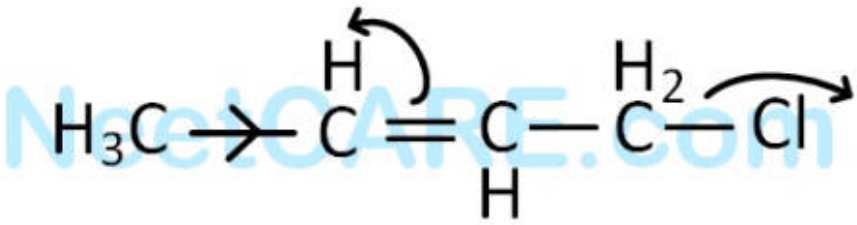
(B)



(C)



(D)



Q54. AIPMT 2015 Cancelled Paper

In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is

Open solution

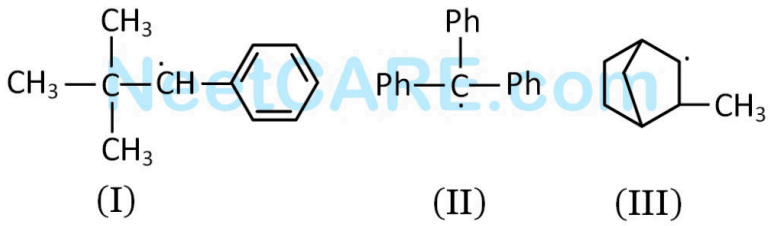
(A) 16.75 (B) 15.76 (C) 17.36 (D) 18.20

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Q55. AIPMT 2015 Cancelled Paper

Open solution

Consider the following compounds :



(I) (II) (III)

Hyperconjugation occurs in

(A) III only (B) I and III (C) I only (D) II only

Q56. AIPMT 2015 Cancelled Paper

Open solution

The enolic form of ethyl acetoacetate as shown below has



(A) 9 sigma bonds and 2 pi-bonds (B) 9 sigma bonds and 1 pi-bond

(C) 18 sigma bonds and 2 pi-bonds (D) 16 sigma bonds and 1 pi-bond.

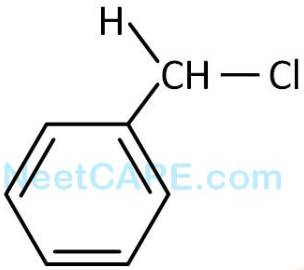
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Q57. AIPMT 2015 Cancelled Paper

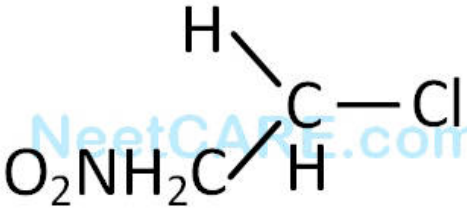
In which of the following compounds, the C—Cl bond ionisation shall give most stable carbonium ion?

Open solution

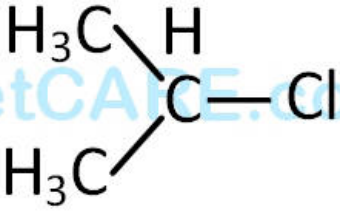
(A)



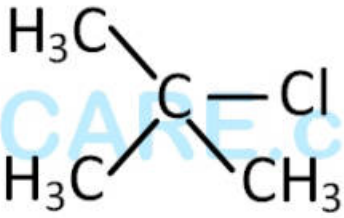
(B)



(C)



(D)



Q58. AIPMT 2014

In the Kjeldahl's method for estimation of nitrogen present in a soil sample, ammonia evolved from 0.75 g of sample neutralized 10 mL of 1 M H_2SO_4. The percentage of nitrogen in the soil is

Open solution

(A) 37.33 (B) 45.33 (C) 35.33 (D) 43.33

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Q59. NEET 2013 (Karnataka)

Open solution

Arrange the following in increasing order of stability

1. $(CH_3)_2 - \overset{+}{C} - CH_2 - CH_3$
2. $(CH_3)_3 - \overset{+}{C}$
3. $(CH_3)_2 - \overset{+}{C}H$
4. $CH_3 - \overset{+}{C}H_2$
5. $\overset{+}{C}H_3$

NeetCARE.com

(A) 5 < 4 < 3 < 1< 2

(B) 4 < 5 < 3 < 1 < 2

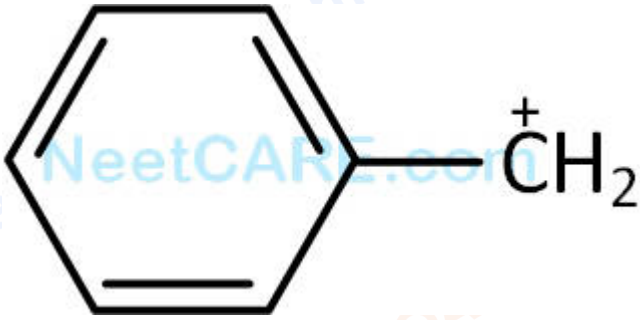
(C) 1 < 5 < 4 < 3 < 2

(D) 5 < 4 < 3 < 2 < 1

Q60. NEET 2013 (Karnataka)

Open solution

What is the hybridisation state of benzyl carbonium ion?



(A) sp^2

(B) spd^2

(C) sp^2d

(D) sp^3

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Q61. NEET 2013 (Karnataka)

Open solution

Homolytic fission of the following alkanes forms free radicals $CH_3 - CH_3$, $CH_3 - CH_2 - CH_3$, $(CH_3)_2CH - CH_3$, $CH_3 - CH_2 - CH(CH_3)_2$.

Increasing order of stability of the radicals is

(A) $(CH_3)_2\dot{C} - CH_2CH_3 < CH_3 - \dot{C}H - CH_3 < CH_3 - \dot{C}H_2 < (CH_3)_3\dot{C}$

(B) $CH_3 - \dot{C}H_2 < CH_3 - \dot{C}HH - CH_3 < (CH_3)_2\dot{C} - CH_2 - CH_3 < (CH_3)_3\dot{C}$

(C) $CH_3 - \dot{C}H_2 < CH_3 - \dot{C}H - CH_3 < (CH_3)_3\dot{C} < (CH_3)_2\dot{C} - CH_2CH_3$

(D) $(CH_3)_3\dot{C} < (CH_3)_2\dot{C} - CH_2CH_3 < CH_3 - \dot{C}H - CH_3 < CH_3 - \dot{C}H_2$

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Q62. NEET 2013 (Karnataka)

Open solution

Which of the following chemical system is non aromatic ?

(A)



(B)



(C)



(D)



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Q63. NEET 2013 (Karnataka)

Open solution

Given :



I and II are

(A) identical

(B) a pair of conformers

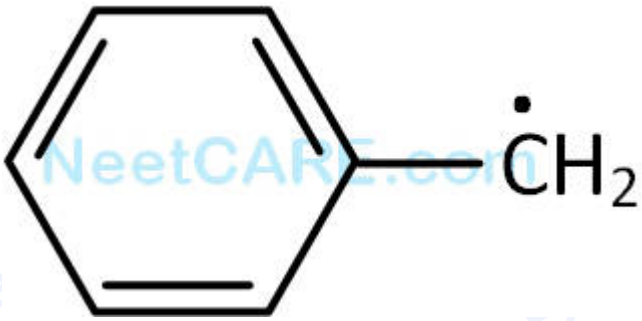
(C) a pair of geometrical isomers

(D) a pair of optical isomers

Q64. NEET 2013

Open solution

The radical,



is aromatic because it has

(A) 7 p-orbitals and 7 unpaired electrons

(B) 6 p-orbitals and 7 unpaired electrons

(C) 6 p-orbitals and 6 unpaired electrons

(D) 7 p-orbitals and 6 unpaired electrons.

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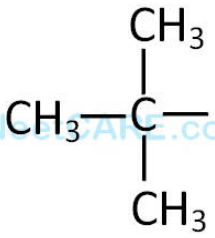
Q65. NEET 2013

The structure of isobutyl group in an organic compound is

(A)



(B)



(C)



(D)



Open solution

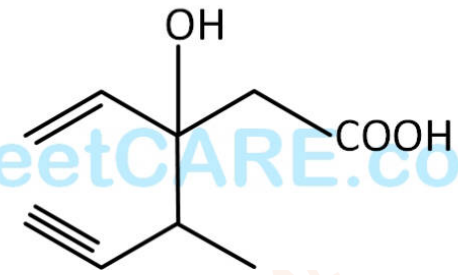
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Q66. NEET 2013

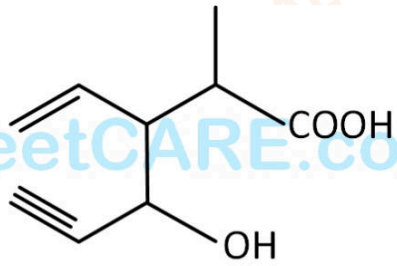
Open solution

Structure of the compound whose IUPAC name is 3-Ethyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid is

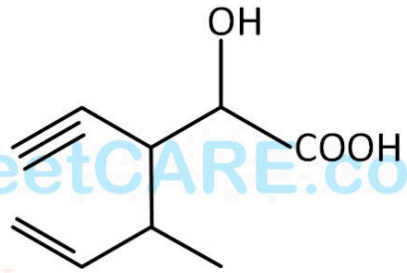
(A)



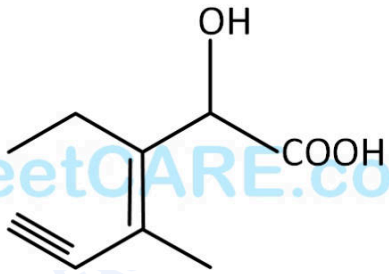
(B)



(C)



(D)



Q67. NEET 2013

Open solution

Some meta-directing substituents in aromatic substitution are given. Which one is most deactivating ?

(A) $-COOH$ (B) $-NO_2$ (C) $-C\equiv N$ (D) $-SO_3H$

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Q68. AIPMT 2012 Prelims

Which nomenclature is not according to IUPAC system?

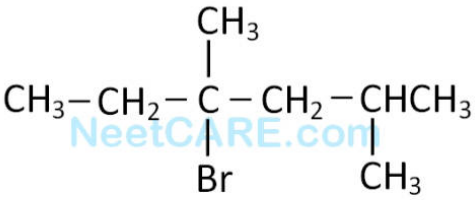
Open solution

(A)



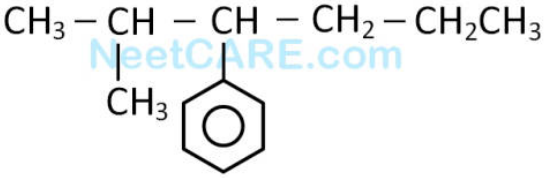
1-Bromoprop-2-ene

(B)



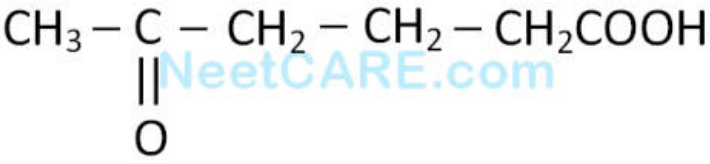
4-Bromo-2,4-dimethylhexane

(C)



2-Methyl-3-phenylpentane

(D)



5-oxohexanoic acid

Q69. AIPMT 2012 Prelims

Which of the following acids does not exhibit optical isomerism?

(A) Maleic acid (B) α-amino acids (C) Lactic acid (D) Tartaric acid

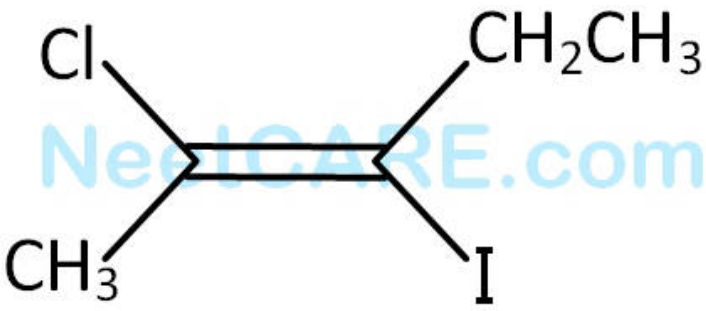
Open solution

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Q70. AIPMT 2011 Mains

The IUPAC name of the following compound is

Open solution



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(A) trans-2-chloro-3-iodo-2-pentene

(B) cis-3-iodo-4-chloro-3-pentane

(C) trans-3-iodo-4-chloro-3-pentene

(D) cis-2-chloro -3-iodo-2-pentene.

Q71. AIPMT 2011 Prelims

Considering the state of hybridization of carbon atoms, find out the molecule among the following which is linear ?

Open solution

(A) $CH_3 - CH = CH - CH_3$

(B) $CH_3 - C \equiv C - CH_3$

(C) $CH_2 = CH - CH_2 - C \equiv CH$

(D) $CH_3 - CH_2 - CH_2 - CH_3$

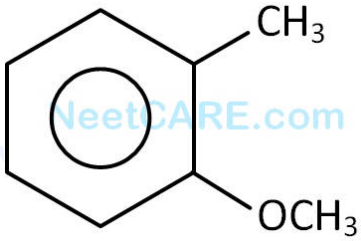
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Q72. AIPMT 2011 Prelims

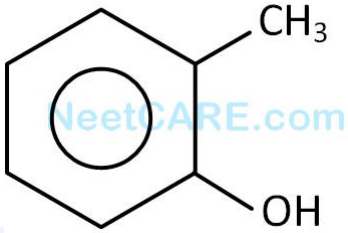
Open solution

Which one of the following is most reactive towards electrophilic reagent?

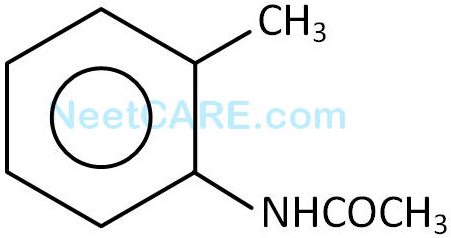
(A)



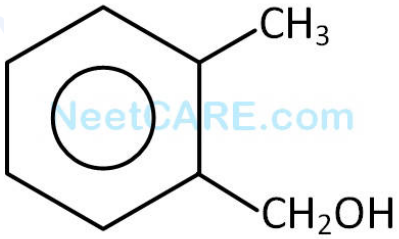
(B)



(C)



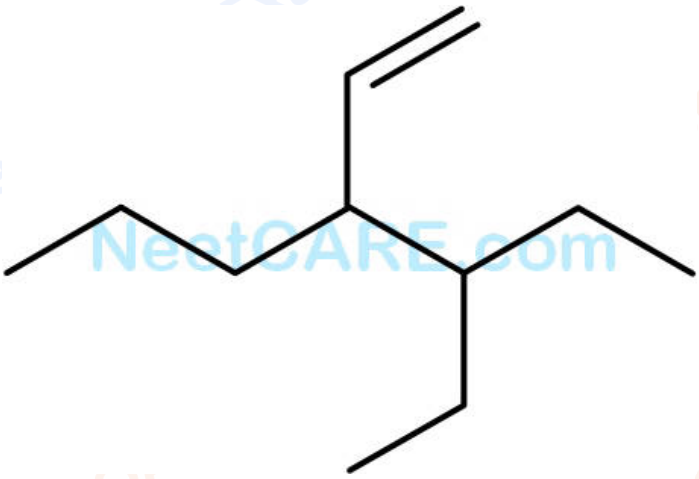
(D)



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Q73. AIPMT 2011 Prelims

The correct IUPAC name for the compound is



Open solution

(A) 4-ethyl-3-propylhex-1-ene
(B) 3-ethyl-4-ethenylheptane
(C) 3-ethyl-4-propylhex-5-ene
(D) 3-(1-ethylpropyl)hex-1-ene

Q74. AIPMT 2010 Mains

Which of the following species is not electrophilic in nature?

(A) $\overset{+}{Cl}$ (B) BH_3 (C) $H_3\overset{+}{O}$ (D) $\overset{+}{N}O_2$

Open solution

Q75. AIPMT 2010 Mains

The IUPAC name of the compound
$CH_3CH{=}CHC{\equiv}CH$ is

(A) pent-4-yn-2-ene (B) pent-3-en-1-yne (C) pent-2-en-4-yne (D) pent-1-yn-3-ene

Open solution

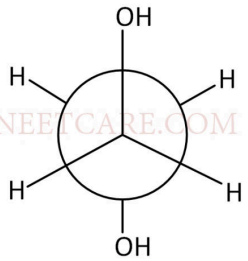
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Q76. AIPMT 2010 Mains

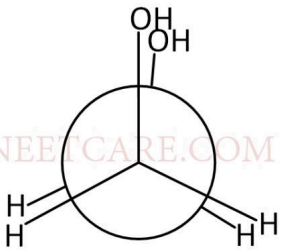
Which of the following conformers for ethylene glycol is most stable?

Open solution

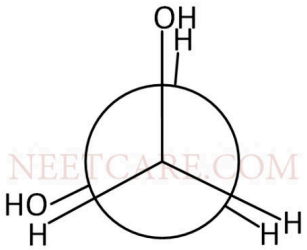
(A)



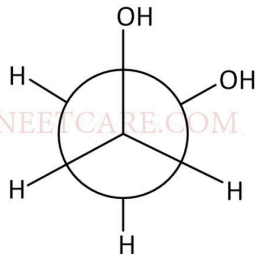
(B)



(C)



(D)



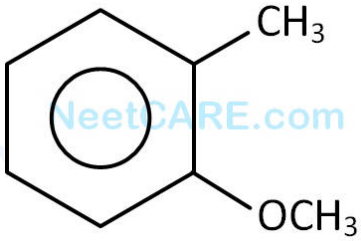
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Q77. AIPMT 2010 Prelims

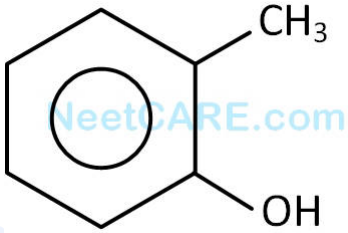
Open solution

Which one of the following is most reactive towards electrophilic reagent?

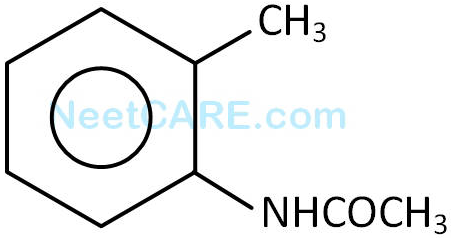
(A)



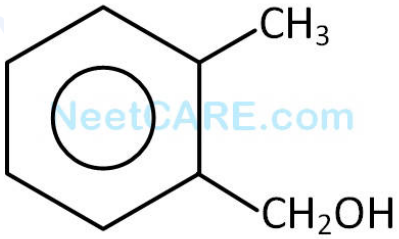
(B)



(C)



(D)

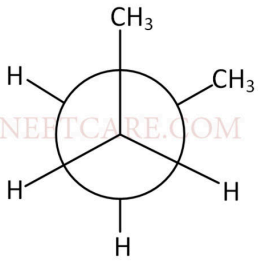


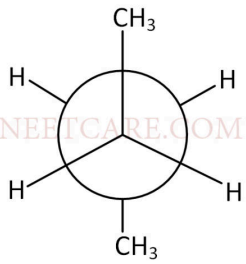
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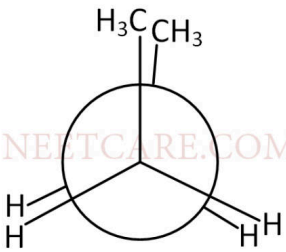
Q78. AIPMT 2010 Prelims

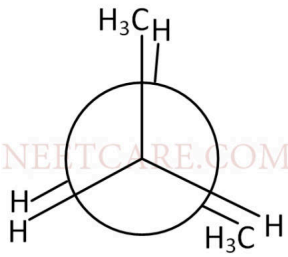
Open solution

In the following the most stable conformation of n-butane is

(A) 

(B) 

(C) 

(D) 

Q79. AIPMT 2009

Open solution

The IUPAC name of the compound having the formula $CH{\equiv}C{-}CH{=}CH_2$ is

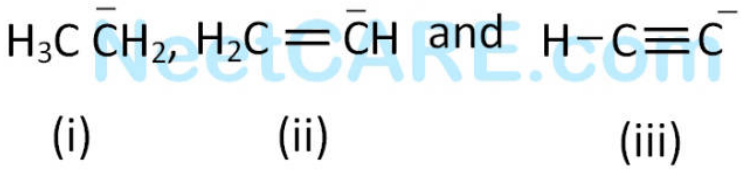
(A) 1-butyne-3-ene (B) but-1-yne-3-ene (C) 1-butene-3-yne (D) 3-butene-1-yne.

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Q80. AIPMT 2008

Open solution

Base strength of



(i) (ii) (iii)

is in the order of

(A) (i) > (iii) > (ii)

(B) (i) > (ii) > (iii)

(C) (ii) > (i) > (iii)

(D) (iii) > (ii) > (i)

NeetCARE.com

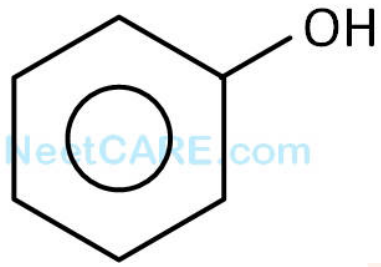
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Q81. AIPMT 2008

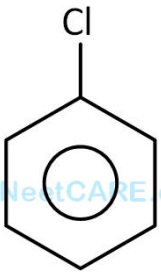
Open solution

Which one of the following is most reactive towards electrophilic attack?

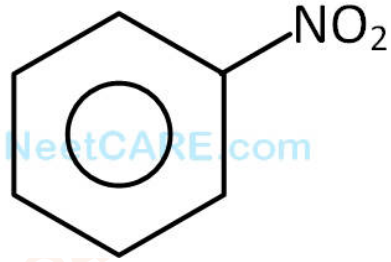
(A)



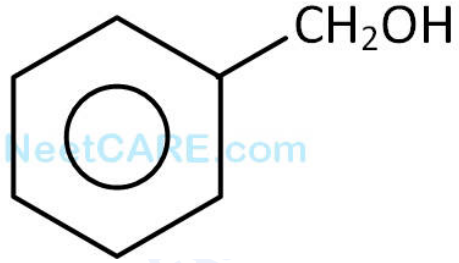
(B)



(C)



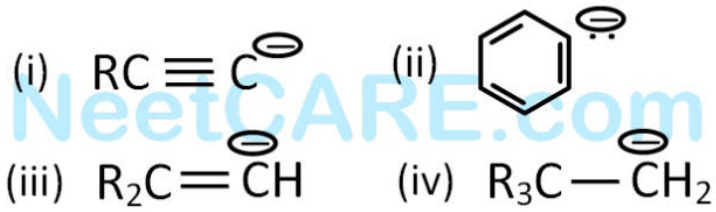
(D)



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Q82. AIPMT 2008

The stability of carbanions in the following.



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is in the order of

(A) (iv) > (ii) > (iii) > (i)
(B) (i) > (iii) > (ii) > (iv)
(C) (i) > (ii) > (iii) > (iv)
(D) (ii) > (iii) > (iv) > (i)

Open solution

Q83. AIPMT 2008

How many stereoisomers does this molecule have?

$CH_3CH{=}CHCH_2CHBrCH_3$

(A) 8 (B) 2 (C) 4 (D) 6

Open solution

Q84. AIPMT 2007

For (i) I^-, (ii) Cl^-, (iii) Br^-, the increasing order of nucleophilicity would be

(A) $Cl^- < Br^- < I^-$ (B) $I^- < Cl^- < Br^-$ (C) $Br^- < Cl^- < I^-$ (D) $I^- < Br^- < Cl^-$

Open solution

Q85. AIPMT 2007

The order of decreasing reactivity towards an electrophilic reagent, for the following would be
(i) benzene
(ii) toluene
(iii) chlorobenzene
(iv) phenol

(A) (ii) > (iv) > (i) > (iii)
(B) (iv) > (iii) > (ii) > (i)
(C) (iv) > (ii) > (i) > (iii)
(D) (i) > (ii) > (iii) > (iv)

Open solution

Q86. AIPMT 2007

If there is no rotation of plane polarised light by a compound in a specific solvent, though to be chiral, it may mean that

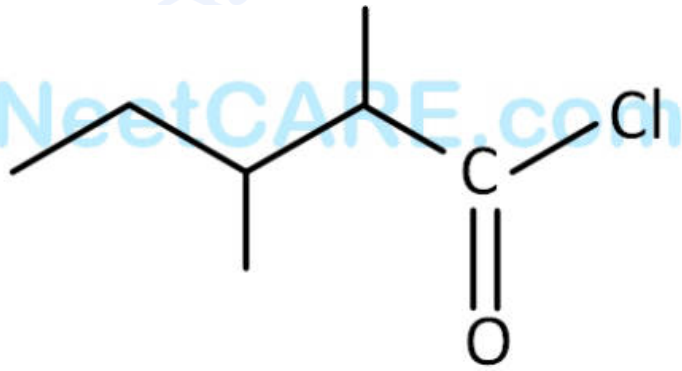
(A) the compound is certainly meso
(B) there is no compound in the solvent
(C) the compound may be a racemic mixture
(D) the compound is certainly a chiral.

Open solution

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Q87. AIPMT 2006

The IUPAC name of



Open solution

is

(A) 1-chloro-1-oxo-2, 3-dimethylpentane

(B) 2-ethyl-3-methylbutanoyl chloride

(C) 2, 3-dimethylpentanoyl chloride

(D) 3, 4-dimethylpentanoyl chloride.

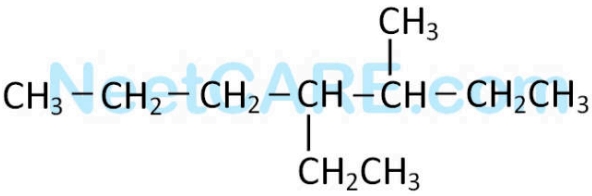
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Q88. AIPMT 2005

Open solution

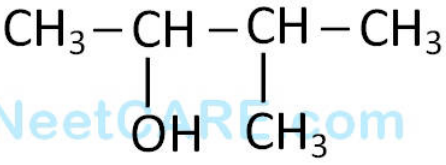
Names of some compounds are given. Which one is not in IUPAC system?

(A)



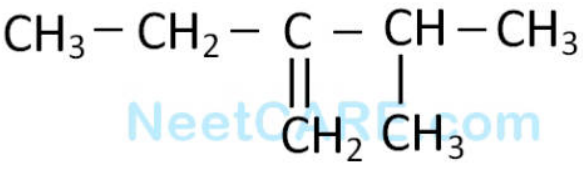
3-Methyl-4-ethylheptane

(B)



3-Methyl-2-butanol

(C)



2-Ethyl-3-methylbut-1-ene

(D)

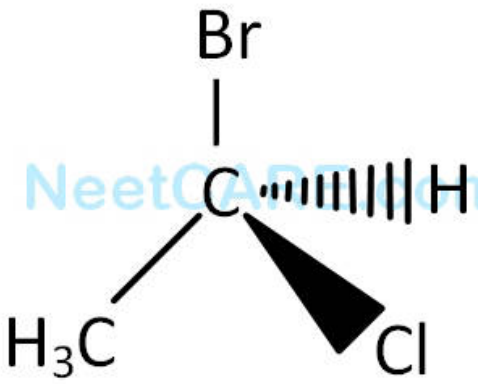


4-Methyl-2-pentyne

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Q89. AIPMT 2005

The chirality of the compound



Open solution

(A) R (B) S (C) E (D) Z

Q90. AIPMT 2005

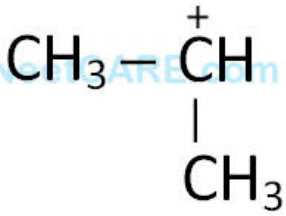
Which amongst the following is the most stable carbocation?

Open solution

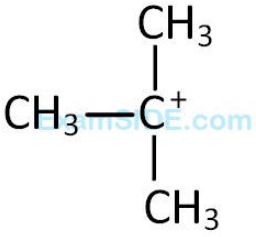
(A) $\overset{+}{C}H_3$

(B) $CH_3\overset{+}{C}H_2$

(C)



(D)



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Q91. AIPMT 2005

Which one of the following pairs represents stereoisomerism?

(A) Structural isomerism and gemetrical isomerism

(B) Optical isomerism and geometrical isomerism

(C) Chain isomerism and rotational isomerism

(D) Linkage isomerism and geometrical isomerism.

Open solution

Q92. AIPMT 2005

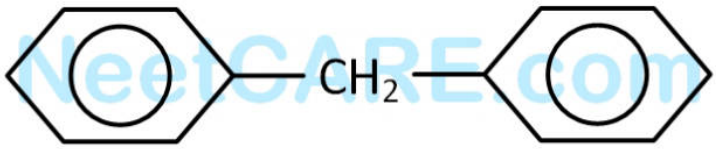
The best method for the separation of naphthalene and benzoic acid from their mixture is

(A) distillation (B) sublimation (C) chromatography (D) crystallisation.

Open solution

Q93. AIPMT 2004

The molecular formula of diphenyl methane,



Open solution

How many structural isomers are possible when one of the hydrogens is replaced by a chlorine atom ?

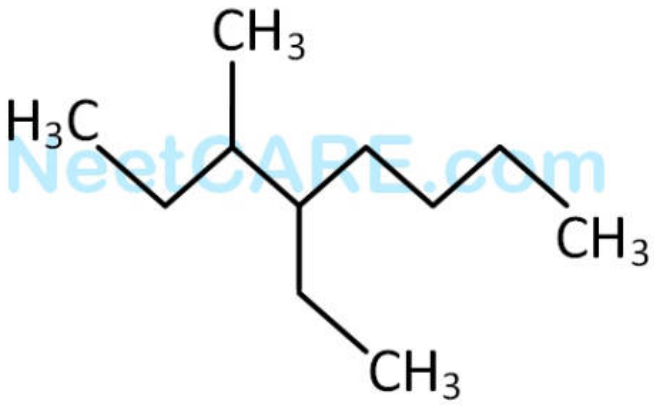
(A) 6 (B) 4 (C) 8 (D) 7

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Q94. AIPMT 2003

Name of the compound given below is

Open solution



(A) 4-ethyl-3-methyloctane
(B) 3-methyl-4-ethyloctane
(C) 2, 3-diethylheptane
(D) 5-ethyl-6-methyloctane

Q95. AIPMT 2003

Which one of the following orders of acid strength is correct ?

Open solution

(A) RCOOH > ROH > HOH > HC ≡ CH
(B) RCOOH > HOH > ROH > HC ≡ CH
(C) RCOOH > HOH > HC ≡ CH > ROH
(D) RCOOH > HC ≡ CH > HOH > ROH

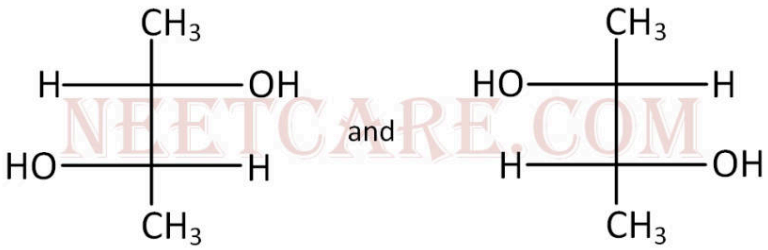
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Q96. AIPMT 2003

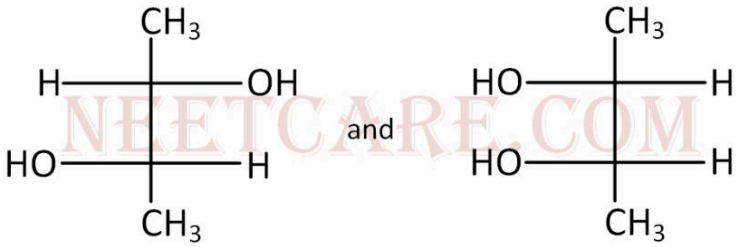
Open solution

Which of the following pairs of compounds are enantiomers?

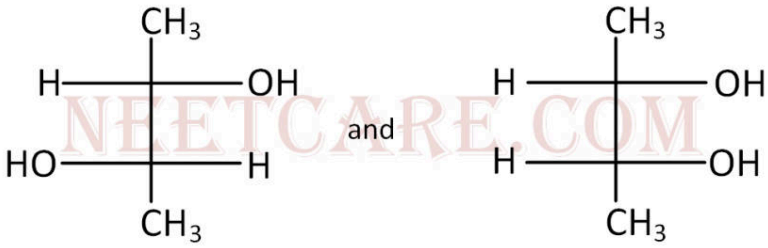
(A)



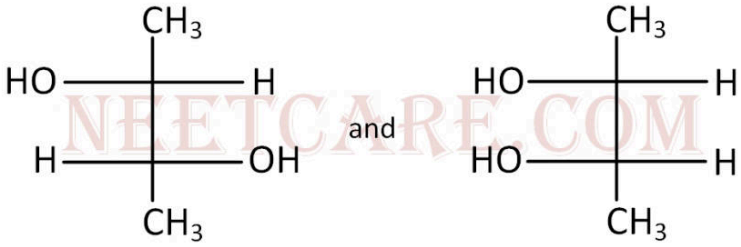
(B)



(C)



(D)



Q97. AIPMT 2002

Open solution

IUPAC name of the following is

$CH_2 = CH - CH_2 - CH_2 - C \equiv CH$

(A) 1-hexyne-5-ene (B) 1, 5 -hexenyne (C) 1-hexene-5-yne (D) 1,5-hexynene.

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Q98. AIPMT 2002

Geometrical isomers differ in

(A) position of functional group

(B) position of atoms

(C) spatial arrangement of atoms

(D) length of carbon chain.

Open solution

Q99. AIPMT 2001

The incorrect IUPAC name is

(A)



2-methyl-3-butanone

(B)



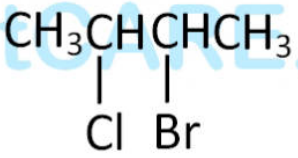
2,3-dimethylpentane

(C)



4-methyl-2-pentyne

(D)



3-bromo-2-chlorobutane.

Open solution

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Q100. AIPMT 2001

In steam distillation of toluene, the pressure of toluene in vapour is

(A) equal to pressure of barometer

(B) Less than pressure of barometer

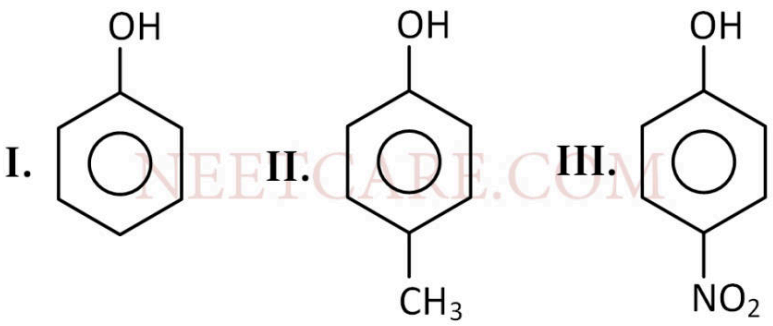
(C) equal to vapour pressure of toluene in simple distillation

(D) more than vapour pressure of toluene in simple distillation.

Open solution

Q101. AIPMT 2001

The correct acidic order of the following is



(A) I > II > III (B) III > I > II (C) II > III > I (D) I > III > II

Open solution

Q102. AIPMT 2001

A compound of molecular formula C_7H_{16} shows optical isomerism, compound will be

(A) 2, 3-dimethylpentane (B) 2, 2-dimethylbutane

(C) 2-methylhexane (D) none of these

Open solution

Q103. AIPMT 2000

The (R)- and (S)-enantiomers of an optically active compound differ in

(A) their solubility in achiral solvents

(B) their reactivity with achiral reagents

(C) their optical rotation of plane polarised light

(D) their melting points

Open solution

Q104. AIPMT 1999

The most suitable method of the separation of a mixture of *ortho* and *para*-nitrophenol mixed in the ratio of 1 : 1 is

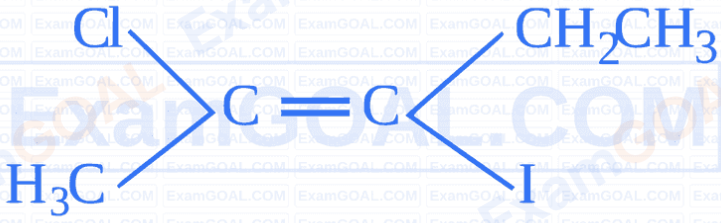
(A) distillation (B) crystallisation (C) vaporisation (D) colour spectrum

Open solution

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Q105. AIPMT 1998

IUPAC name of the compound



(A) *trans*-3-iodo-4-chloro-3-pentene
(B) *cis*-2-chloro-3-iodo-2-pentene
(C) *trans*-2-chloro-3-iodo-2-pentene
(D) *cis*-3-iodo-4-chloro-3-pentene

Open solution

Q106. AIPMT 1998

Which one of the following orders is correct regarding the $-I$ effect of the substituents?

(A) $-NR_2 < -OR > -F$
(B) $-NR_2 > -OR > -F$
(C) $-NR_2 < -OR < -F$
(D) $-NR_2 > -OR < -F$

Open solution

Q107. AIPMT 1998

Which of the following compounds is not chiral?

(A) $DCH_2CH_2CH_2Cl$
(B) CH_3CH_2CHDCl
(C) CH_3CHDCH_2Cl
(D) $CH_3CHClCH_2D$

Open solution

Q108. AIPMT 1997

Tautomerism is exhibited by

(A) R_3CNO_2
(B) RCH_2NO_2
(C) $(CH_3)_3CNO$
(D) $(CH_3)_2NH$

Open solution

Q109. AIPMT 1997

Which of the following technique is most suitable for purification of cyclohexanone from a mixture containing benzoic acid, isoamyl alcohol, cyclohexane and cyclohexanone?

(A) Sublimation
(B) Evaporation
(C) Crystallisation
(D) IR spectroscopy

Open solution

Q110. AIPMT 1996

The number of isomers in $C_4H_{10}O$ will be

(A) 7
(B) 8
(C) 5
(D) 6

Open solution

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Q111. AIPMT 1996

The IUPAC name of $(CH_3)_2CH-CH_2-CH_2Br$ is

(A) 1-bromo-3-methylbutane
(B) 2-methyl-3-bromopropane
(C) 1-bromopentane
(D) 2-methyl-4-bromobutane.

Open solution

Q112. AIPMT 1995

The IUPAC name for

$$CH_3CH=CHCH_2\underset{\underset{NH_2}{|}}{CH}CH_2COOH$$

is

(A) 3-amino-5-heptenoic acid
(B) β-amino-δ-heptenoic acid
(C) 5-amino-2-heptenoic acid
(D) 5-amino-hex-2-enecarboxylic acid.

Open solution

Q113. AIPMT 1995

In the reaction $CH_3CHO + HCN \rightarrow CH_3CH(OH)CN$, a chiral centre is produced. This product is

(A) dextrorotatory
(B) racemic mixture
(C) meso compound
(D) laevorotatory.

Open solution

Q114. AIPMT 1993

The most suitable method of separation of 1 : 1 mixture of *ortho* and *para*-nitrophenols is

(A) chromatography
(B) crystallisation
(C) steam distillation
(D) sublimation.

Open solution

Q115. AIPMT 1992

2-Methyl-2-butene will be represented as

(A) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2CH_3$
(B) $CH_3-\underset{\underset{CH_3}{|}}{C}=CH-CH_3$
(C) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{C}=CH_2$
(D) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH=CH_2$

Open solution

Q116. AIPMT 1992

The IUPAC name of

$$CH_3-\underset{\underset{OH}{|}}{CH}-CH=\underset{\underset{CH_3}{|}}{C}-CHO$$

(A) 4-hydroxy-1-methylpentanal
(B) 4-hydroxy-2-methylpent-2-en-1al
(C) 2-hydroxy-4-methylpent-3-en-5-al
(D) 2-hydroxy-3-methylpent-2-en-5-al

Open solution

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Q117. AIPMT 1992

How many isomers of $C_5H_{11}OH$ will be primary alcohols?

(A) 5 (B) 4 (C) 2 (D) 3

Open solution

Q118. AIPMT 1991

Isomers of a substance must have the same

(A) structural formula (B) physical properties

(C) chemical properties (D) molecular formula.

Open solution

Q119. AIPMT 1991

Which of the following is the most stable carbocation (carbonium ion)?

(A) $CH_3\overset{+}{C}H_2$ (B) $(CH_3)_2\overset{+}{C}H$ (C) $(CH_3)_3\overset{+}{C}$ (D) $C_6H_5\overset{+}{C}H_2$

Open solution

Q120. AIPMT 1991

In sodium fusion test of organic compounds, the nitrogen of the organic compound is converted into

(A) sodamide (B) sodium cyanide (C) sodium nitrite (D) sodium nitrate.

Open solution

Q121. AIPMT 1990

Kjeldahl's method is used in the estimation of

(A) nitrogen (B) halogens (C) sulphur (D) oxygen.

Open solution

Q122. AIPMT 1990

An organic compound *X*(molecular formula $C_6H_7O_2N$) has six carbon atoms in a ring system, two double bonds and a nitro group as substituent, *X* is

(A) homocyclic but not aromatic (B) aromatic but not homocyclic

(C) homocyclic and aromatic (D) heterocyclic and aromatic.

Open solution

Q123. AIPMT 1989

Which one of the following can exhibit *cis-trans* isomerism?

(A) $CH_3-CHCl-COOH$ (B) $H-C\equiv C-Cl$

(C) $ClCH=CHCl$ (D) $ClCH_2-CH_2Cl$

Open solution

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Q124. AIPMT 1989

Lassaigne's test is used in qualitative analysis to detect

(A) nitrogen (B) sulphur (C) chlorine (D) all of these.

Open solution

Q125. AIPMT 1989

Which of the following is an optically active compound?

(A) 1-Butanol (B) 1-Propanol (C) 2-Chlorobutane (D) 4-Hydroxyheptane

Open solution

Q126. AIPMT 1988

How many chain isomers could be obtained from the alkane C_6H_{14}?

(A) Four (B) Five (C) Six (D) Seven

Open solution

MCQ (More than One Correct Answer) 1 questions

Q127. NEET 2018

Which of the following is correct with respect to – I effect of the substituents? (R = alkyl)

(A) $-NH_2 < -OR < -F$ (B) $-NR_2 < -OR < -F$

(C) $-NH_2 > -OR > -F$ (D) $-NR_2 > -OR > -F$

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **A**	3. **B**	4. **B**	5. **A**
6. **C**	7. **C**	8. **B**	9. **D**	10. **A**
11. **D**	12. **D**	13. **C**	14. **A**	15. **D**
16. **C**	17. **B**	18. **D**	19. **B**	20. **D**
21. **D**	22. **B**	23. **B**	24. **C**	25. **B**
26. **D**	27. **D**	28. **D**	29. **B**	30. **A**
31. **D**	32. **C**	33. **C**	34. **A**	35. **A**
36. **A**	37. **B**	38. **A**	39. **C**	40. **D**
41. **D**	42. **C**	43. **B**	44. **C**	45. **C**
46. **D**	47. **A**	48. **B**	49. **D**	50. **B**
51. **D**	52. **B**	53. **A**	54. **A**	55. **A**
56. **C**	57. **D**	58. **A**	59. **A**	60. **A**
61. **B**	62. **D**	63. **B**	64. **C**	65. **C**
66. **D**	67. **B**	68. **A**	69. **A**	70. **A**
71. **B**	72. **B**	73. **A**	74. **C**	75. **B**
76. **D**	77. **B**	78. **B**	79. **C**	80. **B**
81. **A**	82. **C**	83. **C**	84. **A**	85. **C**
86. **A**	87. **C**	88. **A**	89. **A**	90. **D**
91. **B**	92. **B**	93. **B**	94. **A**	95. **B**
96. **A**	97. **C**	98. **C**	99. **A**	100. **B**
101. **B**	102. **A**	103. **C**	104. **A**	105. **C**
106. **C**	107. **A**	108. **B**	109. *Scan QR*	110. **A**
111. **A**	112. **A**	113. **B**	114. **C**	115. **B**
116. **B**	117. **B**	118. **D**	119. **C**	120. **B**
121. **A**	122. **A**	123. **C**	124. **D**	125. **C**
126. **B**				

MCQ (More than One Correct Answer)

127. **A, B**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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